Fixed-Geometry Looped Attention for the Linear Sampling Method

Scope correction and experimental status

August 4, 2026

Abstract

This note corrects the scope of the first proof of concept. The implemented experiment is one-dimensional Gaussian-process regression from direct point samples. It tests a matched fixed softmax/RBF geometry and a tied iterative kernel solver, but it is *not* an experiment on the Linear Sampling Method (LSM). The LSM references instead concern two-dimensional inverse acoustic scattering, complex multistatic or correlation matrices, and a family of regularized linear systems indexed by two-dimensional sampling points. We state the correct source regimes for the three supplied references, formulate the appropriate Bayesian least-squares loop, and specify the experiments required before any result can be described as "ICL for LSM." The existing 1D numbers are retained only as a numerical unit test for the fixed-feature solver mechanism.

1 Scope correction

The previous note used "LSM" as though it denoted a generic least-squares model. In the cited literature it means the **Linear Sampling Method** for inverse scattering. This distinction changes the problem completely. The current experiment samples

$$f \sim \mathcal{GP}(0, k_{\mathrm{RBF}}), \qquad y_i = f(x_i) + \varepsilon_i, \qquad x_i \in [-1, 1],$$

and reconstructs the scalar function f by kernel ridge regression. There is no scattering field, no obstacle, no complex response matrix, and no LSM indicator. It is therefore classical 1D GP regression, exactly as the visual reconstructions suggest.

The correct LSM object is a sound-soft obstacle $D \subset \mathbb{R}^2$. At a fixed wavenumber k, an incident field u^i generates a radiating scattered field u^s satisfying

$$\Delta u^s + k^2 u^s = 0 \quad \text{in } \mathbb{R}^2 \setminus \overline{D}, \qquad u^s = -u^i \quad \text{on } \partial D, \tag{1}$$

together with the Sommerfeld radiation condition. The inverse problem is to recover the geometry of D from near- or far-field measurements.

2 What the Linear Sampling Method solves

In an active far-field experiment with incident plane-wave directions $\widehat{d}_j$ and receiver directions $\widehat{x}_i$, the data are the complex multistatic matrix

$$F_{ij} = u_\infty(\widehat{x}_i, \widehat{d}_j), \qquad F \in \mathbb{C}^{m \times n}. \tag{2}$$

For each probing point $z \in \Omega \subset \mathbb{R}^2$, LSM approximately solves

$$F g_z = \phi_z, \qquad (\phi_z)_i = \frac{e^{i\pi/4}}{\sqrt{8\pi k}} e^{-ik\widehat{x}_i \cdot z}, \tag{3}$$

using SVD/Tikhonov regularization. A standard indicator is

$$I_D(z) = \left(\|g_z\|_2^2 + \tau\right)^{-1/2}. \tag{4}$$

The field $z \mapsto I_D(z)$ is evaluated on a *two-dimensional* probing grid; large values identify the obstacle. With random illumination, the matrix F is replaced by a near-field correlation matrix C or a modified matrix $\widetilde{C}$, but the sampling equation and indicator logic remain.

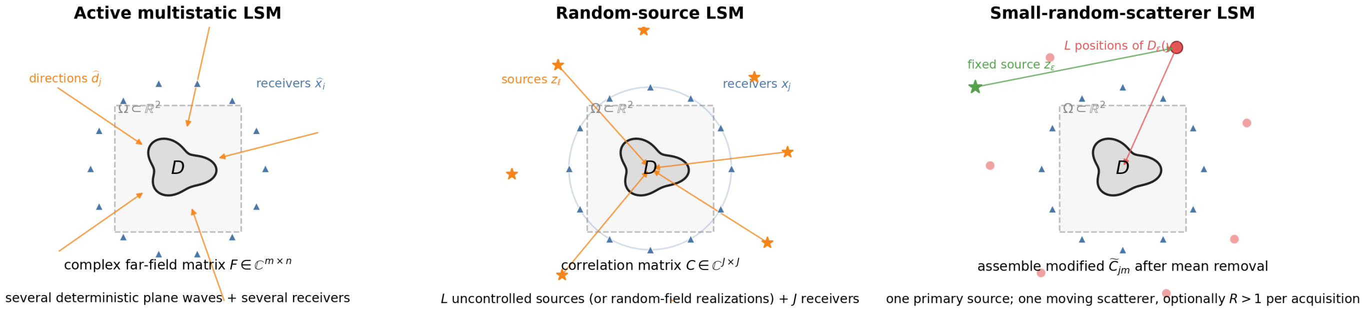

Figure 1: The three acquisition regimes in the supplied references. The random quantity is different in each regime; "random-source LSM" should not be conflated with random GP function samples.

3 One or several sources? Random or deterministic?

The three papers study different acquisition models. The following table is the precise interpretation relevant to a new experiment.

Table 1: Acquisition regimes in the supplied references.

Reference	dimension	illumination/source regime	measured data	reconstruction setting
Garnier–Haddar–Montanelli (2023)	theory $d = 2, 3$; numerics 2D	Several uncontrolled point sources z_ℓ on an enclosing surface; alternatively a continuous random source observed over M realizations.	Receiver cross-correlation $C \in \mathbb{C}^{J\times J}$, approximating the imaginary near-field matrix.	Full and limited aperture; base experiment $J = L = 80$; 100×100 probing grid; ellipse and kite.
Garnier–Haddar–Montanelli (2024)	2D	One fixed, controlled primary point source. One small scatterer changes random position over L acquisitions and acts as a secondary random source. An extension uses $R > 1$ small scatterers per acquisition.	Mean-removed measurements form a modified correlation matrix $\widetilde{C} \in \mathbb{C}^{J\times J}$.	$J = 120$, normally $L = 150$ perturbed positions; $L = 400$ for i.i.d. positions; one or several target obstacles; 100×100 probing grid.
Chenu–Haddar–Montanelli (2026)	2D	Active deterministic full aperture: several plane-wave emitters and several receivers.	Far-field matrix $F \in \mathbb{C}^{m\times n}$.	Fixed RBF-DeepONet trunk; single disks in training, noncircular and two-obstacle tests; LSM validates/refines the neural indicator.

Thus the answer is not simply "one source" or "many sources." In the 2023 paper there are many random primary sources. In the 2024 paper there is one deterministic primary source, while the random moving small scatterer supplies the diversity; several small scatterers are a separate extension. In the 2026 paper the acquisition sources are several deterministic plane waves; the randomness lies in training obstacles and measurement noise, not in the source configuration.

4 Our method: fixed nonlinear GP geometry and looped Bayesian LSM

The references define the physical task; they do not define the proposed learning method. Let A_D be the complex, task-dependent LSM matrix: F_D^δ, C_D^δ, or $\widetilde{C}_D^\delta$. The unknown g_z is indexed by the fixed acquisition nodes ξ_j: incident directions $\widehat{d}_j$ in the far-field setting, or receiver/source locations x_j in the correlation setting. We prescribe a complex GP prior

$$g_z(\xi) \sim \mathcal{CN}(0, \kappa_\Gamma), \qquad (K_\Gamma)_{j\ell} = \kappa_\Gamma(\xi_j, \xi_\ell), \tag{5}$$

on this acquisition geometry. The nonlinearity is chosen from the desired kernel, not learned as a temperature. On a circular array, for example,

$$\kappa_\Gamma(\theta,\theta') = \exp\bigl(\gamma\cos(\theta-\theta')\bigr), \qquad S_{j\ell} = \mathrm{softmax}_\ell\bigl(\gamma\cos(\theta_j-\theta_\ell)\bigr), \tag{6}$$

gives a periodic positive kernel and its row-normalized nonlinear attention. Other Helmholtz- or aperture-adapted kernels may be prescribed in exactly the same way.

With residual covariance Σ and prior covariance $K_g = \tau^2 K_\Gamma$, Bayesian least squares gives

$$g_z^{\mathrm{post}} = \arg\min_g \|A_D g - \phi_z\|^2_{\Sigma^{-1}} + \|g\|^2_{K_g^{-1}} = K_g A_D^*(A_D K_g A_D^* + \Sigma)^{-1}\phi_z. \tag{7}$$

The prior scale τ, residual model Σ, and kernel κ_Γ are modelling choices. They are not outputs of a network. The tied recurrent attention cell solves the dual system in context:

$$\begin{aligned} H_D &= A_D K_g A_D^* + \Sigma, \\ q_z^{(t+1)} &= q_z^{(t)} + \eta_t P_D\bigl(\phi_z - H_D q_z^{(t)}\bigr) + \beta_t\bigl(q_z^{(t)} - q_z^{(t-1)}\bigr), \qquad g_z^{(T)} = K_g A_D^* q_z^{(T)}. \end{aligned} \tag{8}$$

Only the shared solver dynamics—for example a low-dimensional controller for (η_t, β_t) or a structured preconditioner—are trained. The measured operator A_D, the kernel nonlinearity, and the Bayesian covariance model stay explicit.

The posterior covariance

$$\Gamma_D = K_g - K_g A_D^* H_D^{-1} A_D K_g \tag{9}$$

quantifies the uncertainty of the regularized inverse and is useful for acquisition design. A Bayesian LSM indicator can be defined from the posterior expected prior norm,

$$I_D^{\mathrm{Bayes}}(z) = \bigl((g_z^{\mathrm{post}})^* K_g^{-1} g_z^{\mathrm{post}} + \mathrm{tr}(K_g^{-1}\Gamma_D) + \tau_0\bigr)^{-1/2}. \tag{10}$$

For fixed A_D, K_g, and Σ, the trace term is independent of z: it calibrates the indicator magnitude but does not create spatial contrast. The two-dimensional field $z \mapsto I_D^{\mathrm{Bayes}}(z)$ may optionally share a second fixed output kernel $\kappa_\Omega(z,c) = \exp[-\|z-c\|^2/(2\ell_\Omega^2)]$, with $z, c \in \Omega \subset \mathbb{R}^2$. This output geometry is distinct from both the acquisition kernel K_Γ and the task-dependent operator A_D.

Explicit distinction from the 2026 method. Chenu–Haddar–Montanelli learn a preliminary indicator and a noise estimate, combine them into a spatial regularization function, and then run classical LSM. That is *not* the method proposed here. We do not train a DeepONet surrogate of the indicator and do not learn a regularization map. The loop itself performs the Bayesian LSM solve from (A_D, ϕ_z), while the nonlinear kernel geometry is fixed by construction.

5 What the completed 1D experiment establishes—and does not

The existing experiment remains a useful unit test for matched fixed features and tied iteration, but its interpretation is deliberately narrow.

Table 2: Current implementation versus the required LSM experiment.

Component	current experiment	correct LSM experiment
Physical domain	interval $[-1,1]$	probing domain $\Omega \subset \mathbb{R}^2$
Task	fresh scalar RBF-GP draw	fresh obstacle D, source realization, and noise
Observation	direct values $f(x_i) + \varepsilon_i$	complex F_D^δ, C_D^δ, or $\widetilde{C}_D^\delta$
Linear system	$(K+\lambda I)\alpha = y$	$(A_D K_g A_D^* + \Sigma) q_z = \phi_z$
Output	reconstructed scalar function	2D Bayesian LSM indicator from $g_z = K_g A_D^* q_z$
Geometry varied at evaluation	context locations	emitters, receivers, aperture, L/M, k, and 2D query points

At 12 direct samples, the matched 1D loop reaches MSE 0.1205, compared with 0.1136 for exact KRR, 0.1370 for a standard Transformer, and 0.2090 for the mismatched kernel. These numbers support the solver unit test only. They provide no empirical evidence about obstacle localization, LSM regularization, random-source cross-correlations, or inverse scattering.

6 The experiments that must replace it

The correct proof of concept should proceed in three controlled stages.

Stage A: deterministic active LSM. Fix $k = 2\pi$, full-aperture source/receiver circles, $m = n \in \{20, 30, 40, 50\}$, and a 2D probing grid. Generate complex far-field matrices for random disks, ellipses and kites. Train the tied Bayesian loop through the sampling-equation residual and obstacle-level indicator loss, never by imitating an exact linear solve and never by learning the kernel or a spatial regularization map. Evaluate against SVD–Tikhonov–Morozov LSM, with single disks in-distribution and noncircular or multiple obstacles out-of-distribution.

Stage B: random-source LSM. Construct C_D^δ from receiver cross-correlations. Sweep the number of receivers J, random sources L, source perturbation β, number of random field realizations M, noise, and aperture. For i.i.d. random sources or empirical correlations, the reference theory predicts an $O(L^{-1/2})$ or $O(M^{-1/2})$ acquisition error; this is a genuine scaling axis that the learned loop should inherit rather than a context-length law from 1D GP regression.

Stage C: one source with random small scatterers. Fix one controlled primary source, assemble $\widetilde{C}_D^\delta$ from L positions of a small scatterer, and then test $R > 1$ simultaneous small scatterers. The relevant scaling variables are J, L, R, the small scatterer size ε, noise, aperture, and recurrent depth.

For all stages, report relative indicator error, Dice/IoU, boundary Hausdorff or Chamfer distance, localization error, contrast, residual norm, and runtime. Plain latent-function MSE is insufficient. Experimental design must also respect controllability: in active acquisition one may choose incident and receiver directions; in passive random-source acquisition the sources are not controllable, so the design variables are receiver placement, aperture, and acquisition duration/number of realizations.

7 Status

The present code should be labelled *1D fixed-kernel regression control*, not *LSM*. No LSM claim is supported until the input is a complex scattering/correlation matrix and the output is a 2D indicator produced through the sampling equations above. The fixed nonlinearity idea remains relevant, but only as a prescribed 2D feature/prior geometry coupled to the actual task-dependent LSM operator.

References

[1] J. Garnier, H. Haddar, and H. Montanelli. The linear sampling method for random sources. *arXiv:2210.15560*, 2023. `https://arxiv.org/abs/2210.15560`.

[2] J. Garnier, H. Haddar, and H. Montanelli. The linear sampling method for data generated by small random scatterers. *arXiv:2403.19482*, 2024. `https://arxiv.org/abs/2403.19482`.

[3] V. Chenu, H. Haddar, and H. Montanelli. A neural operator framework for solving inverse scattering problems. *arXiv:2602.24147*, 2026. `https://arxiv.org/abs/2602.24147`.